	Algebra Intro
1.	Simplify the following expressions:
a)	$3a - 6b + 8a + 8b$ Ans: $11a + 2b$
b)	$8p + 3q - 5r - 7q + 3p + 2$ Ans: $11p - 4q - 5r + 2$
c)	$2x + 2y + x + (-6y)$ Ans: $3x - 4y$
d)	$3(x + 2) + 2x$ Ans: $5x + 6$
e)	$-(2a + 3b) - a + 8b$ Ans: $5b - 3a$
f)	$6x - 2(x - y) + 5y$ Ans: $4x + 7y$
g)	$-(3x + 4y) - (x + y)$ Ans: $-4x - 5y$
h)	$2(3x - 5) - 3(7 - 4x)$ Ans: $18x - 31$
2.	Simplify the following and express as a single fraction.
a)	$\frac{a+b}{2} + \frac{2a-b}{4}$ Ans: $\frac{4a+b}{4}$
b)	$\frac{2a-3b}{2} - \frac{a-2b}{4}$ Ans: $\frac{3a-4b}{4}$
c)	$\frac{x+3}{5} - \frac{1-x}{2}$ Ans: $\frac{7x+1}{10}$
d)	$\frac{3x-4y}{2} - \frac{2x+y}{4} - \frac{y}{8}$ Ans: $\frac{8x-19y}{8}$
e)	$\frac{a-2b}{3} - \frac{2(x+2y)}{5}$ Ans: $\frac{5a-10b-6x-12y}{15}$

f)	$\frac{3x+4}{10} - \frac{x+7}{15} - \frac{2x-1}{5}$ Ans: $\frac{4-5x}{30}$
3.	Factorise the following.
a)	$4a + 8b$ Ans: $4(a + 2b)$
b)	$4ab - 2b$ Ans: $2b(2a - 1)$
c)	$30x - 5y$ Ans: $5(6x - y)$
d)	$4a + 4$ Ans: $4(a + 1)$
e)	$7xy - 21y$ Ans: $7y(x - 3)$
f)	$18xyz - 6xy$ Ans: $6xy(3z - 1)$
g)	$-50ax + 8x$ Ans: $2x(4 - 25a)$
h)	$-20mx - 3my$ Ans: $-m(20x + 3y)$
i)	$pq - q^2$ Ans: $q(p - q)$
j)	$4x^2 - 3x$ Ans: $x(4x - 3)$
4.	(a) Find the value of $\frac{x-5}{x+7} - 3x^2$ when $x = -2$. Ans: $-13\frac{2}{5}$
	(b) Evaluate $\frac{3a^2bc}{2b-3c} - \frac{bc}{a}$ when $a = 2, b = -3, c = -4$. Ans: 18
5.	Solve the following equations
a)	$9x + 2 = 29$ Ans: $x = 3$
b)	$6 - 4x = 14$ Ans: $x = -2$

c)	$$5x + 4 = 18 - 2x$$ Ans: $x = 2$
d)	$$1 - 3x = x + 13$$ Ans: $x =- 3$
e)	$$2(3x + 4) = 26$$ Ans: $x = 3$
f)	$$2(5 - 2x) = 9$$ Ans:$x = 0.25$
g)	$$2(x - 5) = 3x - 18$$ Ans: $x = 8$
h)	$$6 - 3x = 4(x + 5)$$ Ans: $x =- \frac{2}{7}$
6.	Solve the following equations
a)	$$\frac{2x-1}{3} = \frac{x+2}{4}$$ Ans: $x = 2$
b)	$$\frac{2x}{5} = \frac{x+1}{4}$$ Ans: $x = 1\frac{2}{3}$
c)	$$\frac{1}{4}(3x - 2) = \frac{2}{5}$$ Ans: $x = 1\frac{1}{5}$
d)	$$\frac{3}{x-5} = \frac{6}{x}$$ Ans: $x = 10$
e)	$$\frac{u+5}{6} = \frac{u-2}{3}$$ Ans: $= 9$
f)	$$\frac{a}{a+3} = \frac{3}{4}$$ Ans: $a = 9$